

University of Dhaka  
Affiliated Engineering Colleges  
Department of Computer Science and Engineering  
1<sup>st</sup> Year 1<sup>st</sup> Semester B.Sc. Examination, 2024  
CSE – 1101, Fundamentals of Computers and Computing

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1.
  - a) Explain the basic organization of a digital computer with appropriate figure. 5
  - b) What is a Solid State Drive (SSD)? Describe the working principle of a Hard Disk Drive (HDD). 6
  - c) What happens when you press a key on the computer keyboard? Explain in details. 3
  
2.
  - a) What is the concept of multitasking in an operating system? Explain the difference between system software and application software with examples. 3
  - b) What is a computer network? Differentiate between LAN, MAN, and WAN with examples. 4
  - c) Define MAC address and IP address. Compare the functions of a hub, switch, and router. In which scenarios would each be used? 4
  - d) What is an Extranet? Define URL and describe its structure with an example. 3
  
3.
  - a) What is microprocessor? How does microprocessor work? 4
  - b) Convert the number as following: 5
    - i)  $11001101_2 = ?_8$
    - ii)  $1A7_{16} = ?_{10}$
    - iii)  $5647_8 = ?_{10}$
    - iv)  $817.125_{10} = ?_2$
    - v)  $9DE_{16} = ?_2$
  - c) What is SSD? How an SSD relate to an SDRAM or HDD. 3
  - d) Make comparison between server and workstation. 2
  
4.
  - a) Explain the distinction between data and information with examples. 2
  - b) Define database. What are the advantages of using a Database Management System (DBMS) over a traditional file system for data storage? 4
  - c) What is a domain name? Explain how the Domain Name System (DNS) works to translate domain names into IP addresses. 4
  - d) What is the difference between POP3 and IMAP protocols in email services? How does email spam filtering work? 4
  
5.
  - a) What are the major types of network topologies? Briefly describe with diagram. 6
  - b) Explain the differences between client-server and peer-to-peer network architectures. 4
  - c) Describe LAN, MAN, and WAN with examples. 4
  
6.
  - a) What is program? Write down the basic structure of a C program. 5
  - b) Define flowchart. Write an algorithm to find the largest value among three numbers. 6
  - c) How keywords differs from a variable? One says, float and int are two words that can be used as a variable. Justify your answer. 3
  
7.
  - a) "The main is a part of every C program. C permits different forms of main statement". What are those forms? Explain each of them. 3
  - b) Draw a flow chart with syntax for Nested if....else statement with an example. 4
  - c) Write down the program development cycle. 5
  - d) What will be the output of the following code? 2
 

```
#include <stdio.h>
int main() {
    int a = 5;
    if(a=3) a=7;
    else a=5;
    printf("%d\n", a);
    return 0;}
```